

CUSTOMER CHURN FACTORS AND PREDICTION



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AGENDA

Customer Churn

Initial Data Findings

Residual Analysis

Churn Prediction

Insights

Recommendations



WHAT IS CUSTOMER CHURN?

Customer Churn:

The act of a customer canceling their use of a service or subscription with a business.

Objective:

Understand what makes a customer churn, make recommendations and create a model to predict when a customer shows signs of churning.

Dataset Introduction:

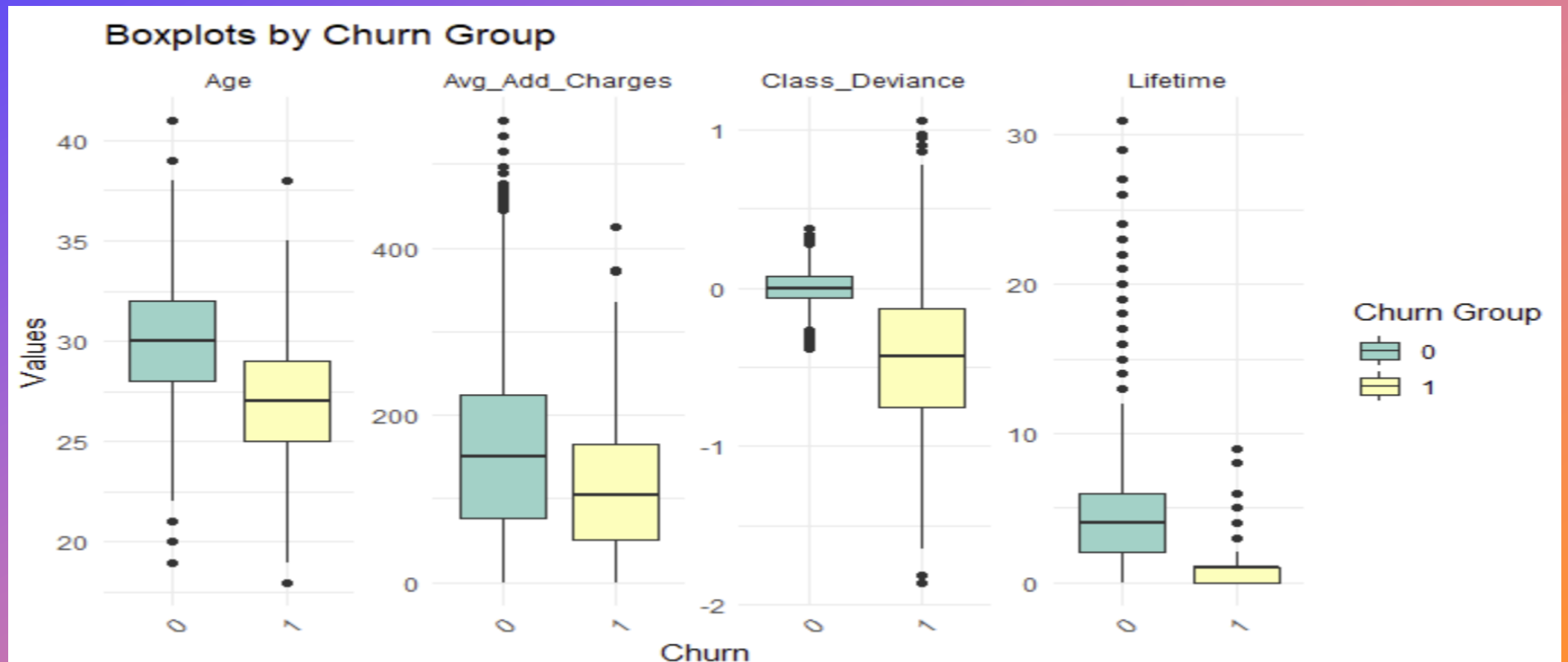
The dataset is on gym customer features and has 4,000 observations.

Variables:

Gender, Near Location, Partner Company, Promo Friends, Phone Number On File, Contract Period, Group Visits, Age, Average additional Charges, Months Till Contract Ends, Lifetime, Average Class Attendance, Current Deviance, Churn

INITIAL FINDINGS

Numerical Summary



INITIAL FINDINGS

Categorical Summary: Proportions by Column

Variable <chr>	Level <chr>	Stay <dbl>	Churn <dbl>
Location	0	0.13	0.23
Location	1	0.87	0.77
Friends_Promo	0	0.65	0.82
Friends_Promo	1	0.35	0.18
Contract_Type	Annual	0.32	0.02
Contract_Type	Monthly	0.43	0.88
Contract_Type	Semi-Annual	0.25	0.10
Bring_Friends	0	0.54	0.73
Bring_Friends	1	0.46	0.27
Variable <chr>	Level <chr>	Stay <dbl>	Churn <dbl>
Class_History	Active	0.51	0.48
Class_History	Beginner	0.32	0.51
Class_History	Regular	0.17	0.01
Contract_Renewal	Later	0.50	0.09
Contract_Renewal	Renew	0.43	0.88
Contract_Renewal	Upcomming	0.07	0.03

INITIAL FINDINGS

Categorical Summary: Proportions by Group

Variable <chr>	Level <chr>	Stay <dbl>	Churn <dbl>
Friends_Promo	0	0.69	0.31
Friends_Promo	1	0.84	0.16
Contract_Type	Annual	0.98	0.02
Contract_Type	Monthly	0.58	0.42
Contract_Type	Semi-Annual	0.88	0.12
Bring_Friends	0	0.67	0.33
Bring_Friends	1	0.83	0.17
Contract_Renewal	Later	0.94	0.06
Contract_Renewal	Renew	0.58	0.42
Contract_Renewal	Upcomming	0.87	0.13
Variable <chr>	Level <chr>	Stay <dbl>	Churn <dbl>
Class_History	Active	0.75	0.25
Class_History	Beginner	0.63	0.37
Class_History	Regular	0.97	0.03

RESIDUAL ANALYSIS

Tests of Homogeneity: Statistical method used to determine whether different populations or groups have the same distribution of a categorical variable.

Table of Categorical Variables and P-values of G-Test with Churn Groups	
Column <chr>	P_Value <dbl>
Gender	9.642976e-01
Location	4.773959e-15
Partner_Company	0.000000e+00
Friends_Promo	0.000000e+00
Contract_Type	0.000000e+00
Bring_Friends	0.000000e+00
Contract_Renewal	0.000000e+00
Class_History	0.000000e+00

RESIDUAL ANALYSIS

Residual Analysis: A method used after a chi-square test to identify which cells in a contingency table contribute most to the overall association or difference between groups.

- **Friends Promo:** customers who enrolled in this promotion were highly overrepresented among those who stayed (residual = 10.2)
- **Contract Type:** customers on Annual and Semi-Annual contracts are overrepresented among those who stayed (residuals = 19.4 and 10.3, respectively), while those on Monthly contracts are heavily underrepresented (residual = -25.1)
- **Bring Friends:** Exhibit an overrepresentation in the "stay" group (residual = 11), indicating that social engagement may play a role in continued membership.
- **Class History:** Active and Regular participants are more likely to stay (residuals = 1.8 and 13.2), while Beginners are underrepresented (residual = -11.1). This suggests that newer or less engaged members may be at greater risk of churn

LOGISTIC REGRESSION MODELS

Name_of_Model <chr>	Accuracy <dbl>	AUC <dbl>	Sensitivity <dbl>	Specificity <dbl>
Lasso	0.9374	0.9753	0.9841	0.8082
Ridge	0.9291	0.9735	0.9807	0.7862
Ridge No Gender	0.9308	0.9737	0.9807	0.7925

3 rows

Lasso is the best model for predicting customer Churn.

- Practical to use the model with the least amount of variance.
- Other model suggestions: KNN, SVM, and Random Forrest



INSIGHTS

Social:

- Offering friend passes and encouraging a supportive environment will lead to the retention of customers.
- Have personal trainers walk around the facility to engage with customers.

Promotional:

- Advertise longer contract periods when customers sign up.
- Advertise signing up to bring guests during onboarding.

Services:

Classes help predict if a customer will churn based on their attendance history and their current monthly usage compared to the average.

- + Send supportive or follow up emails to customers after they attend a class could help retention for those who are at risk.

Retention:

- The first 10 months of a new customer is critical. Planning out phases of promotions, support, and reminders of services could help
- Younger individuals are more likely to churn. Providing classes or personalized messages to them may help retention.
- Promote discounts on products or other services.

RECOMMENDATIONS

Further Analysis

Exploratory Data Analysis

Survey Creation

Customer perspective of gym strengths and areas of improvement

More Data for Analysis

Access to products, other services, foot traffic, equipment, space availability, and facility amenities



QUESTIONS OR FEEDBACK?

THANK YOU!

